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An Improved Quasi-Steady Model Capable of Calculating Flexible Deformation for Bird-Like Flapping Wings

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Abstract

Flapping wing air vehicles (FWAVs) are bio-inspired aerial robots that boast high flight efficiency, excellent maneuverability, and stealth capabilities. However, the dynamic modeling techniques for flapping motion remain underdeveloped, particularly for bird-like FWAVs. Traditional quasi-steady (QS) models' accuracy heavily relies on empirical coefficients, leading to unacceptable errors when analyzing flapping motion at different scales and wing flexibilities. This paper presents an improved QS model that incorporates the interaction mechanism between wing flexible deformation and relative flow state, enabling quantitative analysis of wing deformation's aerodynamic influence and significantly reducing reliance on empirical coefficients. The model comprises two solvers: an aerodynamic solver that outputs wing surface force distribution at each discrete time point, and a structural deformation solver that provides corresponding relative flow state distribution. In each iteration cycle, these solvers exchange their output data until they converge to a stable wing surface displacement distribution and output the predicted time-varying aerodynamic data. Wind tunnel experiments, utilizing high-speed cameras and six-dimensional force sensors, validated the reliability of both the structural deformation solver and the aerodynamic solver. This aerodynamic model, balancing speed and accuracy, shows strong potential for applications in FWAVs' structural and controller design, including but not limited to airfoil optimization and flight attitude stabilization.

Keywords: Flapping wing air vehicles; Dynamic modeling; Aerodynamic; Wind tunnel experiment

1 Introduction

Since the 21st century, the complex mechanisms of flapping flight have gradually been unveiled. With the continuous advancement of microelectromechanical systems (MEMS) and robotics technology, flapping flight has been realized on the scale of small unmanned aerial vehicles (UAVs), giving rise to the novel bio-inspired aerial robots known as FWAVs. Compared to micro rotary and fixed-wing aircraft, FWAVs offer superior maneuverability, efficiency, low noise, and strong stealth capabilities[1,2]. These attributes hold significant potential for applications in environmental monitoring, urban warfare, and military reconnaissance. However, the propulsion of bio-inspired FWAVs relies entirely on wing flapping, and the wing surfaces are flexible structures composed of various mechanical properties materials, resulting in highly complex flow fields during flapping flight. This complexity renders existing dynamic modeling techniques for flapping motion insufficiently mature[3,4], often necessitating extensive wind tunnel experiments to determine the effectiveness of FWAV structures and control strategies, thereby impeding their development.

The flight mechanisms of flapping motion differ significantly from those of traditional propeller-based mechanisms, with propulsion generated by wing flapping at low Reynolds numbers[5,6], characterized by high temporal variability, nonlinearity, and difficulty in quantification[7]. Numerous factors influence the

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aerodynamic performance of flapping motion, including flapping frequency, amplitude, wing mass and stiffness distribution, body motion posture, and flight speed[8–11] Steady-state models based on momentum or actuator disc theory, lifting line theory, or vortex ring theory struggle to accurately describe such complex and variable motion[12]. Unsteady models based on the traditional Kutta-Joukowski condition also face applicability issues when addressing the repeated directional changes in flapping motion[4,13]. In contrast, QS models remain the most commonly used approach for flapping dynamics modeling.

Chin et al.[14] summarized the aerodynamic mechanisms in flapping motion as: (1) leading-edge vortices (LEV), (2) Wagner effect, (3) clap and fling, (4) rotational circulation, (5) wing-wake interaction, and (6) added mass. Based on these aerodynamic mechanisms, Dickinson et al.[15–17] studied fruit fly experimental data and proposed a QS model based on empirical coefficients, considering translational circulation, rotational circulation, wake capture effects, and added mass effect. This QS model, currently the most widely applied, eliminates the need to solve the complex process of unsteady flow, enabling rapid prediction of instantaneous aerodynamics in flapping motion. However, its accuracy is entirely dependent on empirical coefficients[4].

Empirical coefficients derived from experimental methods struggle to accommodate diverse structural and flight parameters. The translational circulation coefficient in Dickinson's model is expressed as a single-variable function of the angle of attack (AOA)[16], failing to account for factors like wing vein layout, wing shape, and wing material. Han et al.[18] improved the expression of the translational circulation coefficient based on the variation of the aerodynamic center with AOA, but the revised empirical coefficients still cannot be universally applied to all types of FWAVs. This empirical data-driven formula remains accurate for micro-scale FWAVs due to typically low disturbance indoor testing environments and flight modes primarily involving hovering or low-speed translation[19–22]. However, for bird-like FWAVs performing high-speed operations in outdoor environments with wind disturbances[23–25], flexible wing deformation significantly alters local flow states. Constructing an aerodynamic model with a wide range of uncertain flight parameters through empirical coefficients is challenging, requiring extensive experimental data and incurring high costs in terms of time and money.

Investigating the impact of flexible wing deformation on aerodynamic characteristics is one of research topics for advanced FWAVs[1]. Heathcote et al.[26] noted in their experiments that flexible wing deformation slightly increases thrust but reduces lift for FWAVs. Pourtakdoust et al.[27] studied the propulsion mechanism of FWAVs in a no-wind, zero-degree absolute AOA environment and developed an aeroelastic model, concluding that active propulsion for bird-like FWAVs is entirely driven by wing flexible deformation. Liu et al.[28] used computational fluid dynamics (CFD) methods to establish a two-dimensional motion simulation model of flexible wings compared to rigid wings, finding that flexible wings significantly enhance FWAV thrust. Qin et al.[29] analyzed the deformation process of flexible wings and wing veins using the Workbench platform, indicating that increased wing flexibility results in increased drag. Quantifying the influence of flexible wing deformation on aerodynamics might be addressed using fluid-structure interaction (FSI) based CFD methods. However, solving the overall FSI equations involves a high-dimensional non-homogeneous system with issues of ill-conditioning and computational intensity[9,30]. Many improved FSI solvers for FWAVs[31–34] ensure numerical stability but remain computationally expensive and require substantial fluid dynamics knowledge for solver creation, limiting widespread application.

Rapidly and accurately describing the distribution of flexible wing deformation and its aerodynamic interactions is a critical challenge. To measure real-time flexible wing deformation, Yuan et al.[35] used depth cameras for dynamic 3D reconstruction of FWAVs. Tay et al.[36] proposed an experimental and data-driven hybrid method, capturing wing deformation during flapping with high-speed cameras and using it as input for numerical solvers, bypassing complex and resource-intensive FSI problems. While effective for micro-scale FWAVs, these methods may not suit bird-like FWAVs in more complex flight environments. Nakata et al.[37] fitted CFD simulation data using least squares and proposed a CFD-informed QS model, validated with hovering hawkmoths, outperforming traditional QS model. However, constructing the initial database still demands significant computational resources.

In this context, our study proposes an improved QS model that considers the interaction mechanism between

wing flexible deformation and relative flow state. Featuring aerodynamic and structural deformation solvers, it iteratively updates parts of the QS model using an implicit step-by-step algorithm, quantitatively describing the impact of flexible deformation on aerodynamics. This improved QS model can compute a set of parameter combinations in seconds, with accuracy validated through experiments using high-speed cameras and six-dimensional force sensors. Balancing speed and accuracy, this modeling method shows strong potential for guiding structural and controller design.

The rest of the article is organized as follows. Section II introduces the modeling procedure, including the principles of the aerodynamics solver, structural deformation solver, and partitioned implicit algorithm. Section III details the wing flexible deformation and aerodynamic characteristics monitoring experiments conducted in a wind tunnel environment to verify the reliability of the modeling method. Section IV designs two kinds of FWAVs prototypes based on the proposed theory and verifies their flight capabilities. Section V summarizes the results and outlines future work.

2 Flexible Deformation-Relative Flow State Interaction Modeling

In the existing research on flapping wing aerodynamics, it is common to combine the QS model with the blade element model (BEM)[37–39]. This approach significantly simplifies the modeling process but introduces the issue of solution accuracy being heavily dependent on empirical coefficients[4]. This problem is particularly pronounced for bird-like FWAVs with larger wing sizes and higher flight speeds, where significant flexible deformation causes notable changes in relative flow conditions. These effects are challenging to describe accurately using empirical coefficients alone. In this section, we design an aerodynamic solver based on the QS model and a structural deformation solver based on the finite element model (FEM). The interaction between wing flexibility and relative flow conditions is then analyzed using a partitioned implicit algorithm.

2.1 Aerodynamic Solver

First, Fig. 1 provides a clear definition of the coordinate systems and angular quantities involved in this study. In Fig. 1(a), the Z-axis of all coordinate systems can be confirmed by the right-hand rule. The body coordinate system is defined as $O_bX_bY_bZ_b$, where O_b coincides with the center of mass of the aircraft, X_b points to the front of the aircraft, and Y_b points to the right of the aircraft. The trajectory coordinate system is defined as $O_tX_tY_tZ_t$, where X_t always aligns with the direction of the aircraft's velocity, and Y_t coincides with Y_b . The wing coordinate system is defined as $O_wX_wY_wZ_w$, where X_w points to the tail of the aircraft and Y_w aligns with the midline of the wing's leading edge. The absolute angle of attack α_0 is the angle between X_t and X_b, while the flapping angle θ is the angle between X_w and Y_b. In Figs. 1(b) to 1(d), the purple area represents the position of the wing in an ideal rigid state, the green area represents the flexible deformation state, and the red triangular area represents a differential element formed by any three nodes on the wing surface. The chordwise twist angle ε_c^e is defined as the complementary angle of the sharp dihedral angle between the plane of the differential element and the $Y_wO_wZ_w$ plane, while the spanwise twist angle ε_s^e is the complementary angle of the sharp dihedral angle between the plane of the differential element and the $X_wO_wZ_w$ plane. The positive directions for both angles are illustrated in Figs. 1(c) and 1(d).

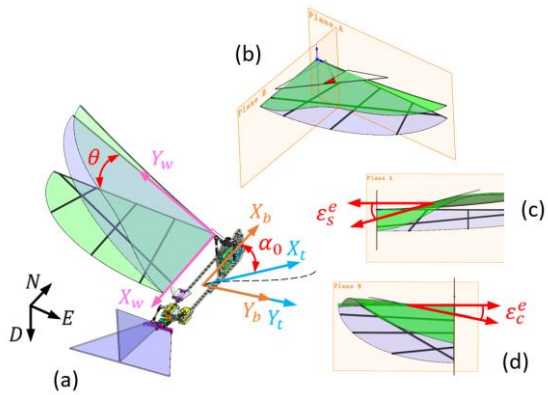

Fig. 1 Definitions of coordinate systems and angles

The core idea of QS model is to simplify unsteady flow problems by decomposing them into a series of instantaneous steady-state problems[40]. This is achieved by discretizing the flapping motion over time and analyzing the aerodynamic characteristics at each discrete phase as a steady state. The instantaneous dynamic equation for each discrete time point is given by:

$$F_{inst} = F_{tran} + F_{rot} + F_{add} + F_{iner} \tag{1}$$

whereF_{tran} is the translational circulatory component, F_{rot} is the rotational circulatory component, F_{add} is the added mass force, and F_{iner} is the inertial force. Since bird-

like FWAVs have low flapping frequencies, special aerodynamic mechanisms such as wake capture and clap-and-fling effects[41] are considered negligible. The calculation method for the circulation component is based on the widely used Dickinson model[16], but it is not established within the traditional BEM framework. To more accurately represent the relative flow conditions, a two-dimensional FEM is used to divide the wing surface into fine grids. The translational circulatory force is computed as:

$$F_{tran} = \frac{1}{2}\rho|v_b + v_e|^2 \int_A C_{tran}^e dS^e \tag{2}$$

where ρ is the fluid density, v_b is the velocity vector in the body coordinate system, v_e is the velocity vector of the external wind, S^e is the effective projected area of the differential element, and C_{tran}^e is the translational circulatory coefficient, determined by the empirical formulas:

$$\begin{cases} C_{L,tran}^e = 1.870\sin(1.872\alpha^e) \\ C_{D,tran}^e = 1.667 - 1.585\cos(1.968\alpha^e) \end{cases} \tag{3}$$

where $C_{L,tran}$ and $C_{D,tran}$ are the lift and drag coefficients of the translational circulation, respectively, and are functions of the element's relative angle of attack, α^e. The physical meaning of α^e is the difference between the absolute angle of attack, α_0, in the trajectory coordinate system and the element chordwise twist angle, ε_c^e, in the wing coordinate system.

The rotational circulation force in Eq. 1 is caused by the passive chordwise twist of the wing membrane, and this effect is particularly significant when the wing membrane area is large and highly flexible[42,43]. It can be computed as:

$$F_{rot} = \frac{1}{2}C_{rot}\rho\omega_\lambda \int_A \dot{\varepsilon_c^e} r_s^e r_c^e dS^e \tag{4}$$

with

$$\omega_\lambda = \frac{A}{2}\frac{\partial}{\partial t}g(f,t) \tag{5}$$

and

$$C_{rot} = 2\pi(0.75 - \hat{x}_0) \tag{6}$$

In Eq. 4, C_{rot} is the rotational circulatory coefficient, ω_λ is the angular velocity of the flapping motion, r_s^e is the spanwise distance from the geometric center of the elemental area to the wing's rotation axis in the wing coordinate system, and r_c^e is the chordwise distance. In Eq. 5, A is the absolute value of the difference between the upper and lower limit angles of a flapping stroke, and $g(f,t)$ represents the periodic function describing the flapping motion gait. Due to mechanical constraints in generating flapping motion for bird-like FWAVs, $g(f,t)$ is typically a sinusoidal function[44,45]. In Eq. 6, $\hat{x}_0$ is the dimensionless position of the flapping motion's pitch axis relative to the leading-edge axis.

The added mass and inertial forces in Eq. 1 are expressed as[17]:

$$F_{add} = \frac{\pi}{4}\rho \int_A {r_c^e}^3 \left[(\dot{\omega}_\lambda sin\alpha^e + \omega_\lambda \dot{\alpha}^e cos\alpha^e) r_s^e - \frac{\ddot{\alpha}^e}{4} r_c^e\right] dS^e \tag{7}$$

$$F_{iner} = \dot{\omega}_\lambda \int_A r_s^e \rho_w^e B_w^e dS^e \tag{8}$$

where ρ_w^e and B_w^e are the density and the thickness of the wing membrane material.

Through coordinate transformation, the thrust force F_T, lift force F_L, and yaw centripetal force F_C can be obtained:

$$\begin{bmatrix} F_T \\ -F_L \\ F_C \end{bmatrix} = R_y(-\alpha_0)R_x(\theta)\begin{bmatrix} 0 \\ 0 \\ F_{inst} \end{bmatrix} \tag{9}$$

where $R_i(\cdot) \in SO(3)$ represents the rotation matrix around the *i* axis.

2.2 Structural Deformation Solver

The wing structure of biomimetic FWAVs mimics that of birds and has been simplified by reducing the complex structures of bones, muscles, and feathers to a planar wing network composed of carbon fiber wing veins and elastic wing membranes. This simplification not only reduces the degrees of freedom in motion but also makes the stiffness distribution of the wing surface more traceable. However, accurately establishing the overall stiffness equation remains challenging due to several factors: (1) The very thin wing membrane lacks bending stiffness and exhibits nonlinear large deformations; (2) There is a complex coupling relationship between the in-plane stress of the wing membrane and the deformation of the wing veins.

Creating a high-precision, nonlinear three-dimensional finite element model is evidently too complex and computationally inefficient, making appropriate simplifications necessary. In fact, for typical bird-like FWAVs, considering factors such as wear resistance and material costs, the wing membrane material usually does not have a very high elastic modulus[46,47]. This implies that the flexible deformation of the wing membrane is continuously and smoothly distributed between the wing veins, allowing the deformation of the entire wing to be

described by the deformation of a few key wing vein nodes. The following assumptions are made for the finite element solution process:

Assumption 1: The displacement of the wing membrane element nodes along the Z_w axis of the wing coordinate system is smoothly distributed between adjacent wing vein element nodes. The transverse displacement in the wing coordinate system is neglected, and the tensile stress of the membrane element is generated entirely by normal displacement-induced deformation.

Assumption 2: The wing vein elements only bear forces along the Z_w axis, including their own inertial forces and the Z_w component of the contact force with the membrane. The contact force at the boundary of the wing veins satisfies a linear distribution relationship.

Based on these assumptions, the flexible deformation of the wing membrane can be roughly estimated by the distribution function between key wing vein nodes without precise calculation. This approach reduces the structural analysis to balancing aerodynamic loads and tensile stresses on the wing vein boundaries. The wing veins, bearing forces in one direction, can be modeled as elastic beams. Fig. 2 shows the finite element mesh layout for a common wing shape, where three-node constant strain triangle (CST) elements[48] are used for the membrane and two-node elastic beam elements[49] for the wing veins. This meshing method ensures that wing vein and membrane nodes coincide, facilitating the application of generalized coordinates and boundary conditions.

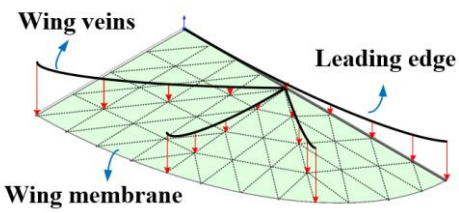

Fig. 2 Example of finite element mesh layout for the wing

For the membrane part, the external load on the *i*-th membrane element, $f_{i,inst}^{e}$, is considered to act on the center point of the element. The external load at the *j*-th node of the membrane element is expressed as:

$$P_{w,j}^{ext} = \sum \frac{f_{i,inst}^{e}}{3}, i = \{k\} \tag{10}$$

where $\{k\}$ denotes all membrane element numbers containing the *j*-th node.

The generalized coordinates $r = [u \;\; v]^T$ describe each membrane region segmented by the wing veins, with the wing vein nodes distributed on the coordinate axes and endpoints $(u_L, 0)$ and $(0, v_L)$. The coordinates of the *j*-th node of the membrane element are (u_j, v_j), and the coordinate transformation is given by:

$$r_j(u_j, v_j) = J_{pr} p_j(x_j, y_j) \tag{11}$$

where $p_j(x_j, y_j)$ is the horizontal and vertical coordinates of the node in the wing coordinate system, and J_{pr} is the Jacobian matrix that maps *p* to *r*.

According to Assumption 1, if the normal displacement of the *j*-th wing membrane element node is h_j, then its in-plane coordinate change is:

$$\begin{cases} u_j' = \left(1 + \dfrac{h_j^2}{u_j^2 + v_j^2}\right)^{\frac{1}{2}} u_j \\ v_j' = \left(1 + \dfrac{h_j^2}{u_j^2 + v_j^2}\right)^{\frac{1}{2}} v_j \end{cases} \tag{12}$$

For a three-node CST element, the shape functions can be written as follows[48]:

$$N_i = \frac{1}{2A}(a_i + b_i u + c_i v), i = 1,2,3 \tag{13}$$

where A is the area of the triangular element, and a_i, b_i, c_i are coordinate transformation constants, which can be solved through the node coordinates. Then, the strain function matrix can be derived as:

$$[B] = \frac{1}{2A}\begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix} \tag{14}$$

The expression for the element stiffness matrix of the wing membrane is:

$$K_w^e = \int_A B^T D B dA \tag{15}$$

where D is the elastic coefficient matrix of the wing membrane material.

The global stiffness matrix $[K_w]$ is assembled by the node order for each wing membrane region. In practical operations, to accelerate the computation, the global stiffness matrix can be defined as a sparse matrix. According to the stiffness equation, the in-plane load of its nodes is obtained as:

$$[P_w^{in}] = [K_w][\Delta r_w] \tag{16}$$

For the *j*-th wing membrane node, its in-plane load component on the Z_w axis of the wing coordinate system is:

$$P_{w,j}^{in,z} = P_{w,j}^{in} h_j (u_j^2 + v_j^2 + h_j^2)^{-\frac{1}{2}} \tag{17}$$

The total load at each wing membrane node is obtained as:

$$P_{w,j} = P_{w,j}^{ext} + P_{w,j}^{in,z} \tag{18}$$

According to Assumption 2, the contact force distribution function at the wing vein boundary is:

$$R_j(u) = \begin{cases} \vartheta_j u + \varrho_j & ,0 \le u \le u_j \\ \dfrac{\vartheta_j u_j}{u_j - u_L}(u - u_L) + \varrho_j & ,u_j < u \le u_L \end{cases} \tag{19}$$

$$R_j(v) = \begin{cases} \varsigma_j v + \varrho_j & ,0 \le v \le v_j \\ \dfrac{\varsigma_j v}{v_j - v_L}(v - v_L) + \varrho_j & ,v_j < v \le v_L \end{cases} \tag{20}$$

where $\vartheta_j, \varrho_j, \varsigma_j$ are undetermined coefficients that can be solved through three equilibrium equations, namely the balance of vertical forces and the moments about two coordinate axes:

$$\begin{cases} \int_0^{u_L} R_j(u)du + \int_0^{v_L} R_j(v)dv = P_{w,j} \\ \int_0^{u_L} uR_j(u)du = P_{w,j}v_j \\ \int_0^{v_L} vR_j(v)dv = P_{w,j}u_j \end{cases} \tag{21}$$

Thus, the load at the i-th wing vein node can be expressed as:

$$P_{b,i} = P_{b,i}^{iner} + \sum R_j(r)l_e \tag{22}$$

where l_e represents the length of each beam element, and $P_{b,i}^{iner}$ represents the inertial force of the wing vein element under the action of flapping motion, can be expressed as:

$$P_{b,i}^{iner} = \rho_c \frac{\pi d_c^2}{4} l_e y_i^2 \frac{\partial \omega_\lambda}{\partial t} \tag{23}$$

where ρ_c is the density of the carbon fiber material, d_c is the diameter of the carbon fiber rod, and y_i is the spanwise coordinate of the midpoint of the beam element in the wing coordinate system.

The wing veins are treated as typical cantilever beam structures, with each node having two degrees of freedom, displacement and rotation. Its element stiffness matrix is[49]:

$$K_b^e = \frac{E_c I_C}{l_e^3} \begin{bmatrix} 12 & 6l_e & -12 & 6l_e \\ 6l_e & 4l_e^2 & -6l_e & 2l_e^2 \\ -12 & -6l_e & 12 & -6l_e \\ 6l_e & 2l_e^2 & -6l_e & 4l_e^2 \end{bmatrix} \tag{24}$$

where E_c is the elastic modulus of the carbon fiber, and I_C is its cross-sectional moment of inertia.

The global stiffness matrix $[K_b]$ for all wing veins is assembled by the node order of the element stiffness matrix K_B^e, and boundary conditions are applied using the penalty method. According to the stiffness equation, the deformation distribution matrix of each wing vein node is obtained as:

$$[q_b] = [K_b]^{-1}[P_b] \tag{25}$$

The normal displacement elements in $[q_b]$ are extracted by the degree of freedom order, and then the normal displacement array $[\Delta]$ of each wing membrane node in the Z_w axis direction is obtained by assigning values using a smooth Fourier function based on the mesh distribution and orientation.

2.3 Partitioned Implicit Algorithm

The computational results from the structural deformation solver are ultimately fed into the QS model to update the relative AOA and the effective projected area of each element. The data exchange process employs a partitioned algorithm, which consists of multiple independent solvers that perform computations separately and then exchange data, significantly improving computational efficiency. To ensure the accuracy of the results, a partitioned implicit algorithm is used[51], involving multiple exchanges and iterative solving within each time step until the results converge to a predefined error limit. The overall data exchange process is illustrated in Fig. 3, where Ω_A^n represents the aerodynamic solver at the n-th discrete time step, and Ω_S^n represents the structural deformation solver.

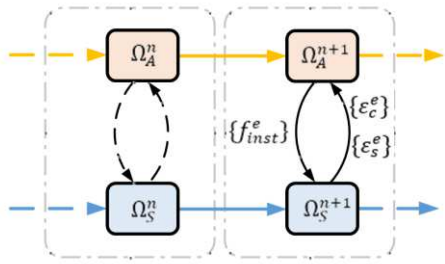

Fig. 3 Flowchart of the partitioned implicit algorithm

The overall methodology is as follows: Initially, the absolute AOA data from the fuselage is used as the initial calculation AOA for each wing membrane element. In the flow field solver, the QS model calculates the instantaneous normal force array $\{f_{inst}^e\}$ for each wing membrane element, which is then input into the structural field solver. Subsequently, the stiffness equation is solved to obtain the normal displacement array $[\Delta]$ for each wing

node. By analyzing the dihedral angle relationship, the chordwise twist angle array $\{\varepsilon_c^e\}$ and the spanwise twist angle array $\{\varepsilon_s^e\}$ for each wing element are derived. The $\{\varepsilon_c^e\}$ and $\{\varepsilon_s^e\}$ arrays can represent the inflow state distribution of the wing surface, where $\{\varepsilon_c^e\}$ updates the aerodynamic coefficients by modifying the relative AOA, and $\{\varepsilon_s^e\}$ reduces the effective projected area of the wing surface, thereby diminishing the effective aerodynamic forces. The updated inflow state is input back into the flow field solver to obtain a new set of instantaneous normal force arrays for the wing elements. This process repeats until the solvers exit the iterative loop and proceed to the next discrete time step.

Specifically, each wing node in the wing coordinate system has three coordinate dimensions: chordwise projection position, spanwise projection position, and normal displacement. The first two are determined and fixed during mesh generation, while the normal displacement changes with each iteration, consequently altering the node's three-dimensional coordinates. Since three-node shell elements are used as the basic elements for the wing surface, each wing element can be regarded as a plane defined by three nodes. The normal vector of this plane can be obtained via the vector cross product:

$$\overrightarrow{v_n^e} = \overrightarrow{p_1^e p_2^e} \times \overrightarrow{p_2^e p_3^e} \tag{26}$$

where $p_i^e = (x_i, y_i)$ represents the coordinates of the counterclockwise ordered nodes of the triangular element.

As shown in Fig. 1(d), the chordwise twist angle ε_c^e and the spanwise twist angle ε_s^e of the element can be derived using the dihedral angle formula:

$$\varepsilon_c^e = \frac{\pi}{2} - arccos\left(\frac{\overrightarrow{v_n^e} \cdot \overrightarrow{g_n}}{\left|\overrightarrow{v_n^e}\right| \cdot |\overrightarrow{g_n}|}\right) \tag{27}$$

$$\varepsilon_s^e = \frac{\pi}{2} - arccos\left(\frac{\overrightarrow{v_n^e} \cdot \overrightarrow{b_n}}{\left|\overrightarrow{v_n^e}\right| \cdot |\overrightarrow{b_n}|}\right) \tag{28}$$

where $\overrightarrow{g_n}$ is the normal vector of the $X_w O_w Z_w$ plane, and $\overrightarrow{b_n}$ is the normal vector of the $Y_w O_w Z_w$ plane.

The convergence criterion for the partitioned implicit algorithm is determined by calculating the difference in the normal displacement array $[\Delta]$ obtained from two consecutive operations. If the absolute value of each element in $[d\Delta]$ is less than the predefined error limit, the iteration loop exits, and the final results are output. Otherwise, bidirectional data exchange continues until the convergence criterion is met. In practice, a relaxation factor is introduced to control the displacement change range of the wing nodes in each computation step, keeping it less than 0.1 times the chord length[34] to prevent solver divergence.

3 Experimental Validation of the Proposed Theory

In this section, we conducted force measurement experiments and motion posture monitoring experiments in a wind tunnel environment to obtain the mechanical data and flexible deformation data of FWAVs under constant-speed cruising conditions. The FWAVs prototypes were successfully manufactured and equipped with various wings of different spans, chord lengths, and wing vein layouts. During the entire experiment, the flight AOA, constant cruising speed, and flapping frequency could be accurately adjusted. By comparing the experimental data with theoretical estimates and analyzing the errors, we determined the effective range of the proposed theory.

3.1 Experimental Platform

The wind tunnel experiments were conducted at the Wind Tunnel and Water Flume of Harbin Institute of Technology (WTWF-HIT). The type of wind tunnel used was a single-return closed-loop wind tunnel, with an error margin of less than 0.1 m/s for wind speeds below 10 m/s. The test section dimensions are 4 m × 3 m × 25 m, which sufficiently meets the requirement of reducing near-wall boundary effects for bird-like FWAVs with wingspans of less than 1 m.

The hardware system configuration is shown in Fig. 4. The flapping motion mechanism is a common spatial four-bar linkage mechanism[45], and the flapping frequency is monitored and controlled in a closed-loop manner using a Hall sensor in the secondary reducer, with a steady-state error of less than 0.1 Hz. The speed control commands are transmitted and received wirelessly. Motion posture detection experiments are conducted using a high-speed camera (VEO-E 310L, Vision Research, Inc.) with a shooting frequency of up to 1000 Hz. Due to layout restrictions in the wind tunnel, the high-speed camera was positioned at a single viewpoint directly in front of the FWAVs prototype. By tracking marked points, all necessary deformation data can be obtained. Force measurement experiments use a six-axis force/torque sensor (FC6D72C, ForceChina, Inc.) with a sampling frequency of up to 1000 Hz and a force resolution of 1/20 N. A fifth-order low-pass Butterworth filter is applied to

the force/torque data, with a cutoff frequency set to three times the flapping frequency to filter out high-frequency vibrations while retaining reliable force data.

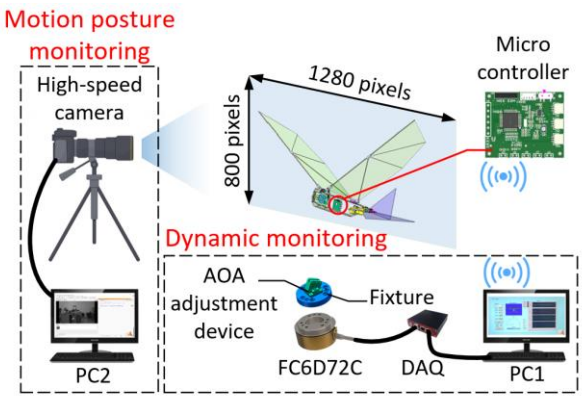

Fig. 4 Hardware system for wind tunnel experiments

3.2 Motion Posture Monitoring Experiment

In previous studies, to accurately describe the flexible deformation of wings during flapping, numerous markers are usually introduced for high-precision motion capture[36], or dynamic 3D reconstruction using depth cameras is directly adopted[35]. However, these approaches are typically due to specific requirements, such as establishing a data-driven dynamic model[36]. In this study, the motion posture detection experiment aims solely to validate the accuracy of the structural deformation solver. Therefore, marking only 4-5 key markers on one side of the symmetric wings is sufficient, focusing solely on their displacement changes in the Z_w axis direction in the wing coordinate system. Fig. 5 shows a frame captured by the high-speed camera, where the origin O_w of the wing coordinate system is the rotation center of the wing leading edge axis, with fixed coordinates. Point P_0 is the connection point at the wing root, aiding in calibrating the flapping angle θ at the current time frame. Point P_1 is the distal end of the wing leading edge axis, while points P_2 and P_3 are the distal ends of the wing veins experiencing significant deformation.

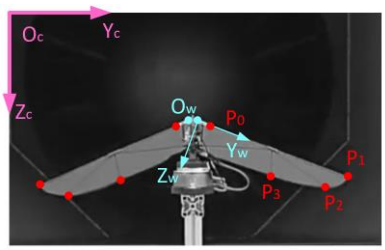

Fig. 5 Markers in the motion posture monitoring experiment

The pixel coordinates of P_1, P_2, P_3 are recorded and then transformed into the wing coordinate system. The coordinate transformation equation from the camera coordinate system to the wing coordinate system is:

$$p_w = k_{cw} R_{cw} (p_c + T_{cw}) \tag{29}$$

where p_w and p_c represent the coordinates of the marker points in the wing coordinate system and the camera coordinate system, respectively. k_{cw} is the projection and scaling matrix, R_{cw} is the rotation matrix, and T_{cw} is the translation matrix, expressed as:

$$k_{cw} = diag\left\{\frac{k_\Lambda}{cos\alpha_0}, \frac{k_\Lambda}{cos\alpha_0}\right\} \tag{30}$$

$$R_{cw} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \tag{31}$$

$$T_{cw} = \left(\overrightarrow{O_c O_w}\right) \tag{32}$$

where k_Λ is the global scaling coefficient, obtained by measuring the pixel vector length between point O_w and marker point P_1 in the static state of the wing with an initial flapping angle of 0 degrees and converting it to the actual wingspan scale.

To validate the accuracy and broad applicability of the improved QS model proposed in this paper, numerous comparative experiments were conducted under different flapping frequencies, different cruising flight speeds, different body absolute angles of attack, and even different wing shapes. Fig. 6 shows the time history of normal deformation data of marker points P_1, P_2, P_3 in the wing coordinate system, where the solid line represents the model prediction data and the scatter points represent the wind tunnel experimental data.

Figs. 6(a) to 6(c) illustrate the effect of flapping frequency on wing flexibility deformation. As the flapping frequency increases, the rotational circulation force increases, leading to greater flexible deformation, without significant attenuation below 8 Hz. When the flapping frequency exceeds 8 Hz, the flexible deformation approaches the structural limit, stabilizing in value. During half-cycle (maximum downward flapping speed), the positive deformation values under a positive installation angle slightly affect the model prediction accuracy due to the small deformation assumption. However, bird-like FWAVs typically do not operate at high flapping frequencies, and the model prediction accuracy at lower frequencies is sufficient.

Figs. 6(d) to 6(f) show the effect of body absolute AOA on wing flexibility deformation under constant-speed

cruising. With a constant flapping frequency and incoming flow speed, a small range change in the AOA does not significantly affect wing flexibility deformation, but an increase in positive AOA causes a shift in overall deformation in the positive direction. This indicates that rotational circulation force is almost unaffected by body AOA, while the translational circulation force positively correlates with body AOA within a small range.

Figs. 6(g) to 6(i) illustrate the effect of incoming flow speed on wing flexibility deformation. As the incoming flow speed increases, the wing flexibility deformation is slightly suppressed, with the most significant suppression near the whole cycle (maximum upward flapping speed). This is due to the opposing directions of translational circulation force and rotational circulation force.

Figs. 6(j) to 6(l) illustrate the effect of different wing stiffness distributions on wing flexibility deformation. The three wing shapes shown have the same wing vein distal endpoints, resulting in identical initial positions of the marker points. Wing type 1 has the lowest spanwise and chordwise stiffness, while wing type 3 has the highest. The deformation of point P_1 reflects the spanwise stiffness, while points P_2 and P_3 reflect the chordwise stiffness. Greater wing flexibility results in greater flexible deformation, especially near the half-cycle when translational and rotational circulation forces align positively, causing peak aerodynamic force and significant deformation.

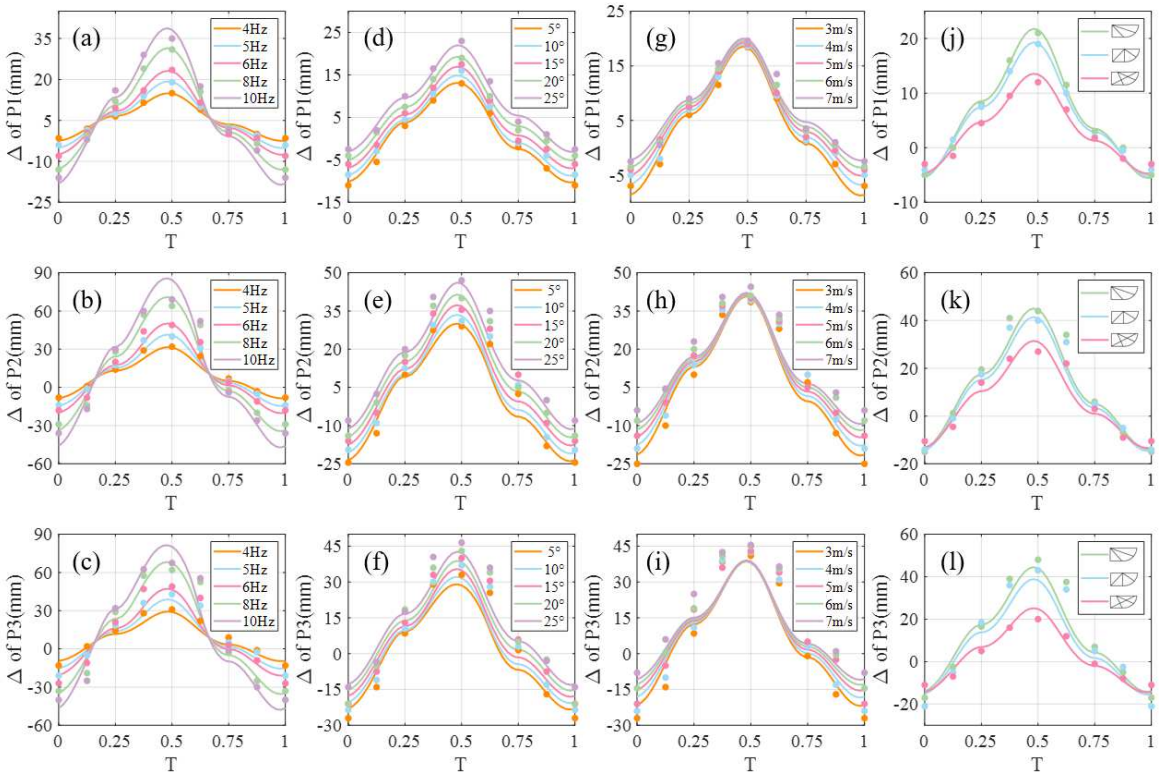

Fig. 6 Time history of flexible deformation over a flapping cycle

Figs. 6(d) to 6(l) were conducted at a flapping frequency of 5 Hz, where the main sources of error are manufacturing inaccuracies, such as dimensional precision and installation gaps. Particularly, the membrane's flatness between the wing veins cannot be guaranteed, with small wrinkles reducing in-plane tensile stress and diminishing its suppression of normal flexible deformation. Additionally, the detachable wing structure causes gaps and rotational issues around the leading-edge axis, affecting points P_2 and P_3 more significantly, especially during downward acceleration (around t = 0.375T) or deceleration (around t = 0.625T). Nonetheless, the improved QS model proposed in this paper still exhibits high prediction accuracy for wing flexibility deformation and broad applicability, reflecting the impact of various design parameters on aerodynamic characteristics. The root mean square error (RMSE) for each set of flight parameters over a flapping cycle is summarized in Fig. 7, showing an overall accuracy within 12 mm.

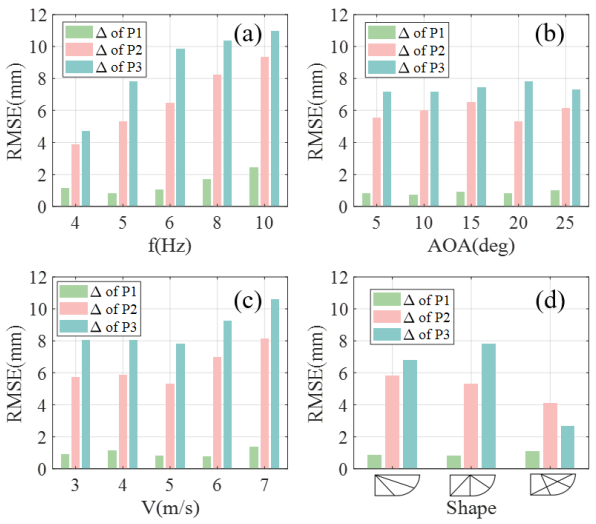

Fig. 7 Root mean square error of flexible deformation over a flapping cycle

3.3 Dynamic Monitoring Experiment

When analyzing the dynamic characteristics of FWAVs, we focus more on the mean data over several stable flapping cycles rather than the large, noisy time history data, as mean data intuitively indicate the flying capability of the designed vehicle. The dynamic data of FWAVs were directly measured using a six-axis force sensor. In the wind tunnel experiment, the trajectory coordinate system and the six-axis force sensor coordinate system overlap due to the constant-speed cruising assumption. Only the forces in the lift and thrust directions are considered, and the mean data are calculated over five complete flapping cycles. Numerous comparative experiments were conducted under different flapping frequencies, body installation angles, incoming flow speeds, and wing shapes. Control groups without wings were used to eliminate the influence of the fuselage and tail on aerodynamic characteristics. The processed wing dynamic data are summarized in Fig. 8, including wind tunnel measured force data, predictions from the improved QS model proposed in this paper, and predictions from the traditional QS model based on BEM. The mean absolute errors (MAE) for lift and thrust data over different parameter groups are summarized in Fig. 9.

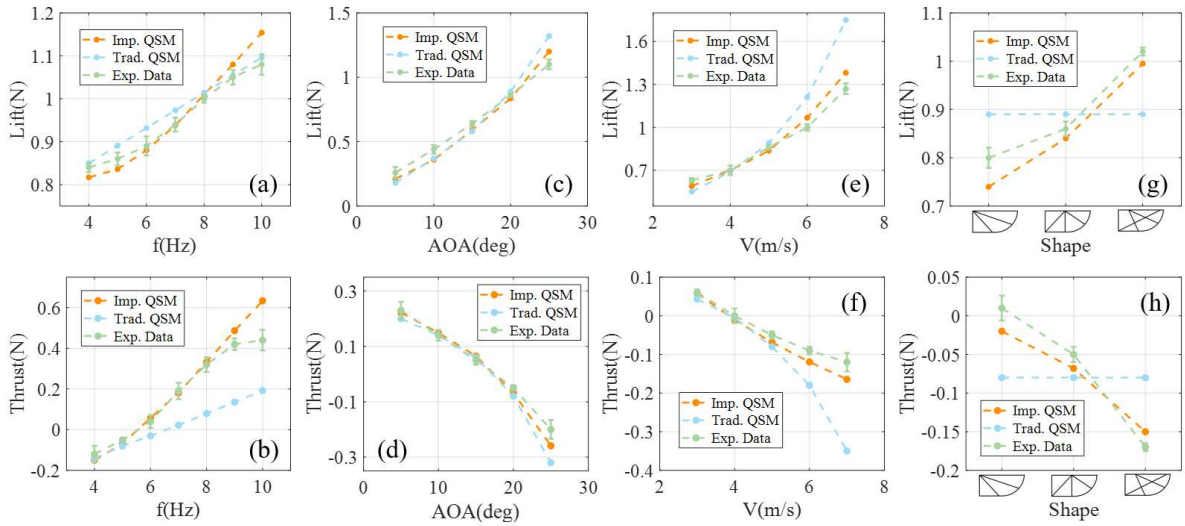

Fig. 8 Mean aerodynamic force data over different parameter groups

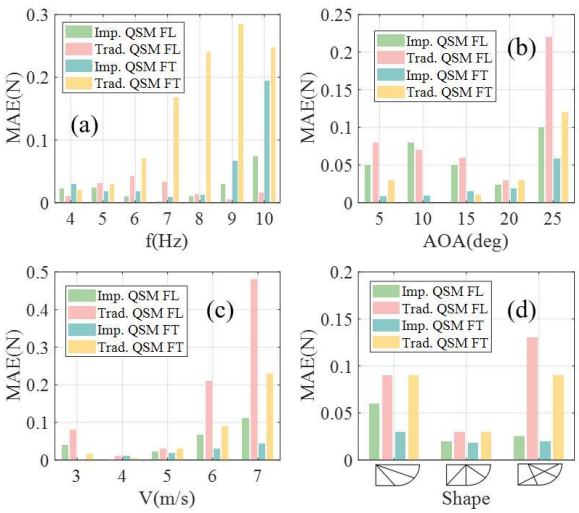

Fig. 9 Mean absolute errors over different parameter groups

Figs. 8(a) and 8(b) show the impact of different flapping frequencies on aerodynamic characteristics. As the flapping frequency increases from 4Hz to 8Hz, the lift data for FWAVs increases by approximately 0.24N, and the thrust data increases by 0.56N. The ratio of these increases is very close to the value of tan(20°), indicating that the flapping frequency primarily affects the rotational circulation force. This force, excluding the periodic oscillating component, forms a propulsive force along the X_b axis in the body coordinate system. This propulsive force can be further orthogonally decomposed into thrust F_T along the X_t axis and lift F_L in the negative Z_t axis direction in the trajectory coordinate system. The improved QS model, due to its ability to quantitatively calculate wing flexible deformation, has high accuracy in solving the rotational circulation force, which translates into high prediction accuracy for lift and thrust. As shown in Fig. 9(a), the prediction accuracy is within 0.05N below 8Hz. However, when the flapping frequency exceeds 9Hz, significant errors occur due to the small deformation assumption. In contrast, the QS model based on BEM estimates the chordwise torsion angle using empirical coefficients, achieving good prediction accuracy only at low frequencies (below 5Hz). Once the actual flapping frequency deviates from this range, the prediction results quickly become distorted.

Figs. 8(c) and 8(d) display the influence of different absolute angles of attack on aerodynamic characteristics. The absolute AOA of the fuselage mainly affects the translational circulation force. The larger the AOA, the greater the components of the translational circulation force in the lift and drag directions. As shown in Fig. 9(b), the improved QS model has better prediction accuracy because the translational circulation force is based on an empirical formula for the AOA, and the BEM-based QS model cannot calculate changes in the AOA due to flexible deformation.

Figs. 8(e) and 8(f) illustrate the effect of different incoming flow speeds on aerodynamic characteristics. Since the translational circulation force has a quadratic relationship with the incoming flow speed, the traditional QS model predicts that lift increases approximately quadratically, while thrust decreases approximately quadratically. However, in reality, as the incoming flow speed increases, the translational circulation force partially cancels out the rotational circulation force, thereby suppressing wing flexible deformation, reducing the relative AOA of the elements, and slowing the growth trend of the translational circulation force. The improved QS model accurately predicts this relationship. As shown in Fig. 9(c), it maintains good prediction accuracy under high wind speed conditions, whereas the traditional QS model becomes distorted.

Figs. 8(g) and 8(h) depict the impact of different wing stiffness distributions on aerodynamic characteristics. The greater the wing stiffness, the smaller the flexible deformation, resulting in an increase in translational circulation force and a decrease in rotational circulation force, reflected as an increase in lift and a decrease in thrust. In the actual design process of FWAVs, the impact of wing flexibility must be fully considered. If the flexibility is too large, sufficient lift cannot be obtained, whereas if the flexibility is too small, sufficient thrust may not be generated. Appropriate flexibility is needed to achieve better aerodynamic characteristics. The improved QS model can accurately predict this relationship, whereas the traditional QS model, driven by empirical formulas, cannot quantitatively calculate wing flexible deformation and, therefore, cannot distinguish the aerodynamic characteristics of different wing shapes.

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In summary, the overall prediction error of the improved QS model for aerodynamic lift is below 0.12N, with approximately 75% of predictions having an error of less than 0.05N. For aerodynamic thrust predictions at flapping frequencies below 10Hz, the error is maintained within 0.07N. In contrast, the traditional QS model only has low error within its applicable empirical coefficient range, with prediction errors reaching close to 0.5N under high incoming flow speeds or high flapping frequencies, making the data entirely unreliable. Furthermore, the improved QS model can analyze the impact of different wing shapes on aerodynamic characteristics, which the traditional QS model cannot achieve. This high-precision, improved QS model has significant application prospects in guiding the structural and controller design of FWAVs.

4 Application of the Proposed Theory

Due to material limitations, the FWAVs' prototype structures, movement patterns, and sensing systems cannot fully mimic real birds, resulting in much lower flight efficiency compared to real birds. The structural parameters designed using the bionic scaling law[52] are often unsuitable. Generally, the structural design of FWAVs requires extensive experimental data to support it. By comparing and analyzing the effects of different angles of attack, incoming flow speeds, flapping frequencies, wing areas, and aspect ratios on aerodynamic characteristics, a suitable set of flight parameters can gradually be found. However, the monetary cost of wind tunnel experiments is prohibitively high, and the time cost of CFD simulation methods is also extremely high. The improved QS model proposed in this study balances low computational cost with high accuracy. Based on this improved QS model, we have developed various bird-mimicking FWAVs based on continuous rotation mechanisms (Fig. 10(a), Prototype 1) and micro servo-driven mechanisms (Fig. 10(b), Prototype 2), achieving stable flapping flight with open-loop control at various flapping frequencies and wingspans, fully demonstrating the model's application value.

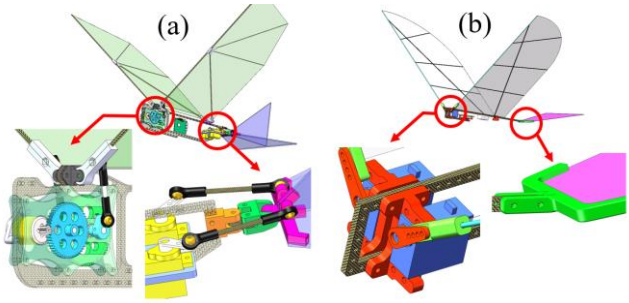

Fig. 10 3D Design Models of Two FWAV Prototypes

These two FWAVs are designed for different operational scenarios. Prototype 1 has excellent power output capabilities, can carry a higher load, and has a longer endurance. Prototype 2 can fully control wing angular position and angular velocity, achieving three-axis attitude control with just two servo motors[46,47,53]. It has better bionic effects, lighter weight, and better maneuverability. This means Prototype 1 has a higher body weight and requires greater aerodynamic lift but can flap its wings faster, while Prototype 2 is the opposite. The improved QS model can simulate the aerodynamic characteristics of both design targets well, aiding in the structural parameter design tasks.

The specific design steps can be divided into the following phases:

Step 1: Propose Initial Design Parameters. We aim to design a medium-sized bird-mimicking FWAVs (with a wingspan between 0.5m and 1m). This size balances load-carrying capacity and flexibility, capable of mounting various small sensors like IMUs, GPS, barometers, and optical flow modules. The smaller turning radius reduces the requirements for the test site. Using 3D modeling software for mass property assessment, the approximate weight of the prototypes carrying various sensors and communication modules is estimated to be around 120g (for Prototype 1) and 80g (for Prototype 2). The wing's aspect ratio is also one of the initial design parameters. In Section 3, we did not study the effect of aspect ratio on aerodynamic characteristics because this parameter needs to consider many non-aerodynamic factors in actual design tasks. Generally, a larger aspect ratio improves aerodynamic characteristics but usually comes with poor

passability and flexibility issues. More critically, increasing the aspect ratio also increases the body's rotational inertia and aerodynamic moment arm, requiring more control and driving torque. Considering all factors, an aspect ratio of 4-6 is acceptable in practical design tasks. In this case, the value is set to 4.

Step 2: Determine Pending Design Parameters. For Prototype 1, the gearbox reduction ratio can be freely configured, meaning the upper limit of the flapping frequency is adjustable. Therefore, the flapping frequency is set as a pending design parameter, while the wing span is set as the initial design parameter, which is 0.6m in this case. Prototype 2 is limited by the output capacity of commercially available micro servo motors, with a maximum continuous bidirectional rotation frequency of about 4Hz. Therefore, the wingspan is set as the pending design parameter.

Step 3: Calculate Pending Design Parameters. Generally, small planar wing FWAVs need a thrust-to-weight ratio of at least 0.4 during takeoff to ensure smooth takeoff and some wind resistance. The proposed improved QS model can accurately calculate the rotational circulation force even in zero wind speed and zero AOA conditions, ensuring high accuracy in aerodynamic thrust calculations. Fig. 11 shows the aerodynamic characteristics as the pending design parameters gradually increase, with the shaded area representing the feasible range where the thrust-to-weight ratio is greater than 0.4. As shown in Fig. 11(a), Prototype 1 needs to achieve a flapping frequency of at least 6.5Hz. As shown in Fig. 11(b), Prototype 2 needs a wingspan of at least 850mm.

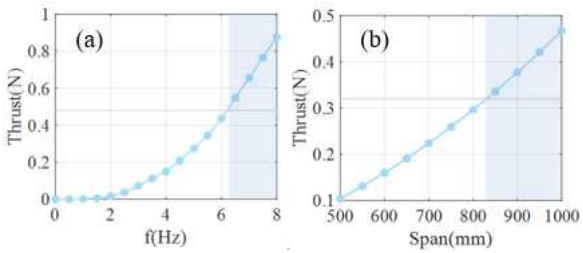

Fig. 11 Aerodynamic Characteristics with Varying Pending Design Parameters

Step 4: Validate Feasible Flight Parameter Space. FWAVs prototypes need to meet two necessary conditions during takeoff and climb: (1) lift greater than weight, and (2) thrust greater than zero. Only increasing flapping frequency or wing area can simultaneously increase lift and thrust. Changing the AOA or flight speed can only increase one while decreasing the other. Therefore, it is necessary to verify whether the designed structural parameters have a feasible range in the flight parameter space. By traversing the AOA and flight speed within common parameter ranges, the flight parameter space is obtained as shown in Fig. 12. Figs. 12(a) to 12(c) show the model data for Prototype 1, and Figs. 12(d) to (f) show the model data for Prototype 2. Figs. 12(a) and 12(d) show the aerodynamic lift surface and prototype weight plane, with the part of the surface above the plane enabling the FWAVs to climb. Figs. 12(b) and 12(e) show the aerodynamic thrust surface and zero drag plane, with the part of the surface above the plane enabling the FWAVs to accelerate forward. The overlapping area in the top view of the extracted Figs. 12(c) and 12(f)indicates the feasible region. The width of about 2-3° in the AOA axis is considered reasonable. If the overlapping area is too small or non-existent, it means the FWAVs prototype has strict flight environment requirements and is challenging to control. If the overlapping area is too large, it means the designed driving performance is excessive and may not match suitable driving components.

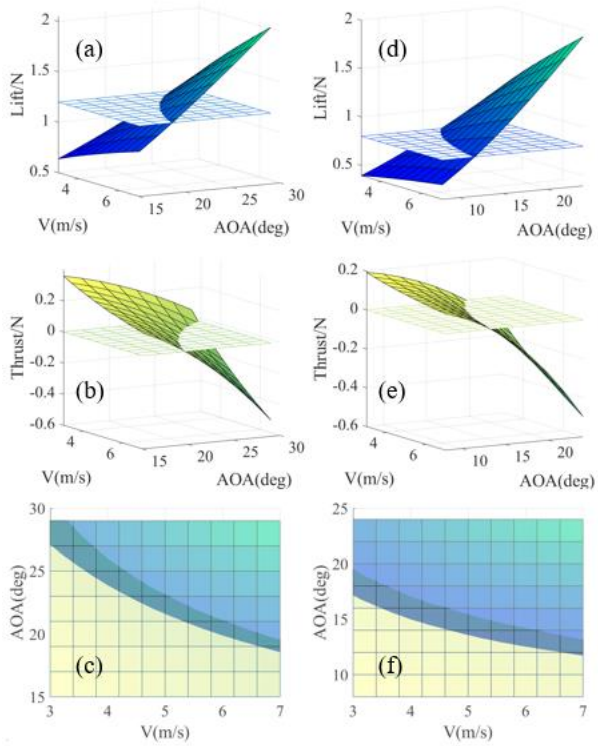

Fig. 12 Verification of Feasible Flight Parameter Space

Step 5: Selection, Design, and Verification of Key Components. Choose the drive motor model for Prototype 1 as EMAX-RS1108, and the designed two-stage gearbox reduction ratio is about 61. The servo motor model for Prototype 2 is GDW-DS290MG.

Step 6: Flight Experiment Verification. Flight experiments of the two designed FWAVs prototypes were conducted in various environments. Fig. 13(a) shows Prototype 1 flying in an indoor corridor, Fig. 13(b) shows Prototype 1 flying in an open area, Fig. 13(c) shows Prototype 2 flying in a forest, and Fig. 13(d) shows Prototype 2 flying between buildings. All flight experiments achieved stable flight with open-loop control, verifying the structural reliability of the designed FWAVs prototypes.

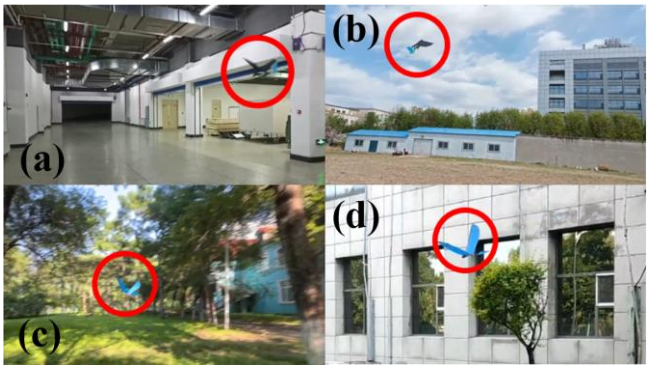

Fig. 13 Verification of Indoor and Outdoor Flight Capabilities

Conclusions

This study proposes an improved QS model capable of quantitatively calculating the flexible deformation of wings, reducing the dependence of the dynamic model on empirical coefficients. The proposed theory has been validated through motion attitude monitoring experiments and dynamic monitoring experiments. Within the common flight parameter range, the model's calculation error for aerodynamic forces is within 0.07N. Additionally, based on the proposed theory, two FWAV prototypes with different drive mechanisms were designed. The effectiveness of the designed structures and the reliability of the design methods were validated through indoor and outdoor flight experiments. In the future, we will continue to explore the application of the improved QS model in further optimizing structural design and guiding controller design to meet the demands of tasks in adverse weather conditions or environments with complex obstacles.

Conflicts of Interest

The authors declared that they have no conflicts of interest.

Data Availability

No data was used for the research described in the article.

Acknowledgments

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